

Certified Short-Cycle Diagnostics for Deployed LDPC Codes: From WiFi to Quantum, with a Covering-Tower Factorization

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Abstract—Short cycles in the Tanner graph of a low-density parity-check (LDPC) code degrade belief-propagation (BP) decoding, and counting them is a standard step in code design and evaluation. This paper does not count them faster. It contributes a short-cycle diagnostic whose defining identity is a *machine-checked theorem*—the trace-formula gap law $\text{tr } A^k - p_k = \text{tr } B^k$, with A the adjacency matrix, B the Hashimoto non-backtracking operator and p_k the power sums of the matching-polynomial roots—and whose per-code outputs are validated by three mutually independent computations that agree exactly. We report a certified census of the four deployed LDPC codes of the IEEE 802.11n (WiFi) standard at block length $n = 648$, and, to our knowledge the first such diagnostic on a *quantum* LDPC code, of the IBM “gross code” [[144, 12, 12]] bivariate-bicycle (BB) code, finding girth 6 and $c_6 = 144$ shortest cycles. We then give a covering-tower factorization: a BB code whose Tanner graph is a double cover of a smaller code’s inherits its short-cycle profile through the classical Artin–Ihara / 2-lift identity $\text{tr } B^k(\tilde{G}) = \text{tr } B^k(G) + \text{tr } B_s^k(G)$, verified exactly on the gross/[[72, 12, 6]] pair (c_6 doubles, $144 = 2 \cdot 72$). The contribution is the theorem-backed, cross-validated, reproducible methodology and its transfer from classical to quantum codes; the underlying mathematics is classical and is not claimed as new. We are explicit about what the certificate does and does not guarantee.

Index Terms—LDPC codes, quantum LDPC codes, Tanner graph, girth, short-cycle enumeration, belief propagation, non-backtracking operator, bivariate bicycle codes, formal verification, reproducible computation.

I. INTRODUCTION

LOW-DENSITY parity-check codes are decoded by belief propagation on a bipartite *Tanner graph* whose two sides are the code’s bits (or qubits) and its parity checks [1], [2]. BP is exact on a tree and only approximate on a graph with cycles: a short cycle lets a message reinforce itself, so the *girth* of the Tanner graph and the *number* of cycles at and just above the girth are quantities a code designer measures [3]. With quantum LDPC (qLDPC) codes now a leading route to fault-tolerant quantum computation [5], the same diagnostic applies to their Tanner graphs.

Counting short cycles is a solved problem: spectral and message-passing algorithms count them efficiently and, on quasi-cyclic graphs, well past the girth [3], [4]. This paper

adds nothing to their speed or reach. It contributes three things of a different kind.

(1) *A theorem-backed, cross-validated census.* The counting identity we use—the trace-formula gap law (Section II)—is a theorem with a complete machine-checked proof in the Lean 4 proof assistant [10], and each per-code count is produced by three computations that share no code path and agree exactly. To the best of our knowledge no prior cycle-counting tool carries a machine-checked guarantee on its defining identity.

(2) *The first such diagnostic, to our knowledge, on a deployed quantum code.* We run the census, unchanged, on the IBM gross code [[144, 12, 12]] [5] (Section IV).

(3) *A covering-tower factorization.* Bivariate-bicycle codes form covering towers [6]; we show the short-cycle profile factorizes across a cover by the classical 2-lift identity, so the profile of a code family is determined by its base (Section V).

We claim no new mathematics: the gap law, the BB covering construction, and the 2-lift factorization are due to others. Section VI states plainly what the certificate guarantees and what it does not, including its relation to the machine-checked *code-distance* certificates of recent formal-verification work [7].

II. THE CERTIFIED CENSUS METHODOLOGY

Let G be the Tanner graph of a code, A its adjacency matrix and g its girth. The *matching polynomial* is $\mu_G(x) = \sum_{j \geq 0} (-1)^j m_j x^{n-2j}$ with m_j the number of j -edge matchings, and $p_k = \sum_i \theta_i^k$ the power sums of its (real) roots. The Hashimoto non-backtracking operator B acts on the $2|E|$ directed edges. The companion theory paper [10] proves, with a `sorry-free` machine-checked proof depending only on the three standard axioms, the *trace-formula gap law*: for every $1 \leq k \leq g + 1$,

$$\text{tr } A^k - p_k = \text{tr } B^k. \quad (1)$$

Below the girth both sides vanish; at $k = g$ both equal $2g c_g$ with c_g the number of shortest cycles, so

$$c_g = \frac{\text{tr } A^g - p_g}{2g} = \frac{\text{tr } B^g}{2g}. \quad (2)$$

Two consequences are used as diagnostics. First, the below-girth vanishing $\text{tr } B^k = 0$ for $k < g$ is a free integrity check on a parsed parity-check matrix: a graph of girth g must report it, and any input failing it is mis-parsed. Second, (2) is the shortest-cycle count.

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The cycle censuses were computed with AI assistance (Claude, Anthropic); the mathematics, the underlying theorem, and all claims are the author’s responsibility. The machine-checked identity is established in the open-source companion development cited herein.

TABLE I

CERTIFIED SHORT-CYCLE CENSUS, IEEE 802.11N LDPC CODES, $n = 648$, $z = 27$. THE THREE INDEPENDENT ROUTES AGREE EXACTLY ON EACH ENTRY.

rate	$ V $	$ E $	girth g	c_g (R1=R2=R3)
1/2	972	2376	6	3 942 hexagons
2/3	864	2376	6	8 046 hexagons
3/4	810	2376	4	54 squares
5/6	756	2376	6	32 346 hexagons

TABLE II

CERTIFIED SHORT-CYCLE CENSUS, H_X TANNER GRAPHS OF THE GROSS CODE AND ITS BASE.

code	$ V $	$ E $	girth g	c_g (R1=R2=R3)
gross $[[144, 12, 12]]$	216	432	6	144
$[[72, 12, 6]]$ (base)	108	216	6	72

Three independent routes. Because a certified *identity* evaluated by fallible code is only as trustworthy as the evaluation, each c_g is computed three ways that meet only at the answer: **R1**, the non-backtracking trace $\text{tr } B^g/2g$ (sparse dart matrix); **R2**, the gap law $(\text{tr } A^g - p_g)/2g$ with p_g obtained by Newton's identities from small matching counts; and **R3**, direct enumeration (common-neighbor counting for $g = 4$, a VF2 subgraph search for $g = 6$). Equation (1) is exactly the theorem that R1 and R2 must agree, so a three-way match also re-confirms the certified identity on each real graph.

III. CLASSICAL: THE IEEE 802.11N LDPC CODES

The IEEE 802.11n standard specifies LDPC codes at three block lengths; we take $n = 648$ with quasi-cyclic lifting size $z = 27$, in all four code rates. Each parity-check matrix yields a Tanner graph on $|V| = n+m$ vertices ($m = n(1-r)$ checks) with a constant $|E| = 2376$ edges. The certified census, with the below-girth zeros confirmed on every code, is Table I.

The four numbers tell a design trade-off: at rate 1/2 the graph is sparse enough to hold girth 6 with the fewest hexagons; at rate 3/4 the density forces the girth down to 4 (54 squares, the most damaging cycles); at rate 5/6 the designers recover girth 6 but pay with 32 346 hexagons. The certificate does not judge the trade; it makes the price exact and reproducible.

IV. QUANTUM: THE GROSS CODE

Nothing in the methodology is classical-channel specific; it takes any Tanner graph. We feed it the IBM gross code $[[144, 12, 12]]$, a bivariate-bicycle (BB) code [5] with $\ell = 12$, $m = 6$, $x = S_\ell \otimes I_m$, $y = I_\ell \otimes S_m$, $A = x^3 + y + y^2$, $B = y^3 + x + x^2$, and $H_X = [A \mid B]$ (the H_X and H_Z Tanner graphs are isomorphic). The census (Table II) finds girth 6 and $c_6 = 144$ shortest cycles, with all three routes agreeing and the Lean-certified below-girth vanishing holding on the quantum code.

V. THE COVERING-TOWER FACTORIZATION

The clean value $c_6 = 2 \cdot 72$ is not a coincidence. BB codes form *covering towers* [6]: the gross Tanner graph is

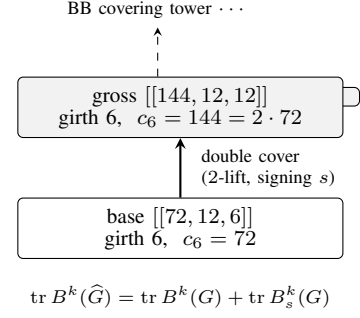


Fig. 1. The covering-tower factorization. The gross code's Tanner graph is a double cover (2-lift) of the $[[72, 12, 6]]$ graph (two sheets, shown stacked); its certified short-cycle profile lifts through the signed non-backtracking trace (3). Girth cycles carry voltage +1, so c_6 doubles ($72 \rightarrow 144$); past the girth the signing is nontrivial (Table III).

TABLE III

VERIFICATION OF THE 2-LIFT FACTORIZATION (3) ON THE GROSS/ $[[72, 12, 6]]$ PAIR, IN EXACT ARITHMETIC. THE IDENTITY HOLDS FOR EVERY k ; PAST THE GIRTH ($k = 8$) THE SIGNED TRACE IS NONTRIVIAL, SO THE FACTORIZATION IS INFORMATIVE, NOT A MERE DOUBLING.

k	$\text{tr } B^k(\hat{G})$	$\text{tr } B^k(G)$	$\text{tr } B_s^k(G)$
1–5	0	0	0
6	1 728	864	864
7	0	0	0
8	27 648	31 104	−3 456

a *double cover* (2-lift) of the $[[72, 12, 6]]$ Tanner graph, which we confirm independently (each base edge lifts to exactly one sheet; $|V(\hat{G})| = 2|V(G)|$). For a double cover with edge signing s , the non-backtracking spectrum splits by the classical Artin–Ihara / 2-lift identity [8], [9]:

$$\text{tr } B^k(\hat{G}) = \text{tr } B^k(G) + \text{tr } B_s^k(G), \quad (3)$$

where B_s is the *signed* non-backtracking operator of the base. We verify (3) exactly for all $k = 1, \dots, 8$ on the gross/ $[[72, 12, 6]]$ pair (Table III). At the girth, $\text{tr } B^6(\hat{G}) = 1728 = 864 + 864$: every base 6-cycle carries voltage +1 and lifts to two cover 6-cycles, so c_6 doubles ($72 \rightarrow 144$). Past the girth the signing is nontrivial—at $k = 8$ the signed trace is -3456 —so the cover's longer-cycle structure genuinely differs from its base. Consequently the certified short-cycle profile of a BB *code family* is determined by its base together with the covering signing, through the Artin–Ihara L -function factorization (Fig. 1)—a practical handle on whole towers rather than one code at a time.

VI. DISCUSSION: SCOPE AND GUARANTEES

We are deliberately exact about what is and is not certified.

The certified object is the identity, not each number. Equation (1) is machine-checked for *all* finite graphs [10]. A per-code count such as $c_6 = 144$ is a computation; its trustworthiness comes from three independent routes agreeing and from the theorem-mandated below-girth vanishing being observed, not from a per-code kernel certificate. We note plainly that neither a cross-check nor a proof assistant validates

that the input graph is the intended code—that is a provenance question—so we do not describe the per-code counts as “kernel-checked.”

No new mathematics. The gap law is classical and folklore-adjacent; the BB covering construction is [6]; the 2-lift factorization is classical [8], [9]. The contribution is the methodology (a theorem-backed identity, cross-validated outputs, reproducible open scripts) and its transfer to deployed classical and quantum codes.

Relation to distance certificates. The load-bearing certificate of a quantum code’s strength is its *distance*, machine-checked by Lean-QEC [7] via a verified SAT reduction. The present diagnostic is decoder-facing and complementary: distance bounds what the code can correct in principle; the short-cycle profile concerns the practical BP decoder.

What it does not reach. The error-floor structures past the girth (trapping and absorbing sets) are not addressed; we count shortest cycles, the entry-level diagnostic.

VII. OPEN QUESTIONS

We leave four questions, in increasing reach, for others to take up.

- 1) *The corrected window.* The gap law is exact for $k \leq g+1$; engineers count to $2g-2$ with correction terms [4]. What walk family does the $k = g+2$ defect $\text{tr } A^k - p_k - \text{tr } B^k$ count, and can the corrected multiplicities be given a theorem-backed form?
- 2) *The tower as a design principle.* The short-cycle profile factorizes across a BB cover through the signed trace (3). Can the covering signing be *chosen* to suppress the short-cycle population—or raise the girth—at every level of the tower, turning the factorization from a diagnostic into a construction rule?
- 3) *From cycles to expansion.* The decoder-relevant quantity for expander-based qLDPC codes is the spectral gap (the Ramanujan property), set by the leading non-backtracking eigenvalue. How tightly do the certified low moments $\text{tr } B^g$ and $\text{tr } B^{g+1}$ bound that gap, and hence the BP threshold?
- 4) *Beyond shortest cycles.* The error floor is governed by trapping and absorbing sets, not by the girth alone. Is there a trace-formula-style identity that certifies the smallest such sets, extending the present diagnostic from the entry level to the structures that actually limit performance?

VIII. CONCLUSION

We have presented a short-cycle diagnostic for LDPC Tanner graphs whose defining identity is a machine-checked theorem and whose outputs are validated by three independent routes, applied to the deployed IEEE 802.11n codes and, for the first time to our knowledge, to a deployed quantum LDPC code, with a covering-tower factorization that lifts the profile across a BB code family. The same pipeline applies to the 3GPP 5G NR base graphs and to larger BB towers; certifying spectral expansion (the Ramanujan property), which governs distance and decoding for expander-based qLDPC codes, is

an active research direction to which the formalized non-backtracking and matching-polynomial machinery used here is naturally suited.

REPRODUCIBILITY

The code constructions, the three-route census and the double-cover factorization check are open at <https://github.com/karlesmarin/godsil-gutman-lean> (research/qldpc-gross/ and the IEEE 802.11n census), together with the Lean 4 sources that prove identity (1).

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